LEGAL PROTECTION AGAINST RETALIATORY FIRING IMPROVES WORKPLACE SAFETY

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Abstract—Workplace safety policies are designed to ensure that employers internalize the costs of injuries, but employers can undermine these policies with threats of dismissal. We show that states' adoption of the public policy exception to at-will employment—an exception forbidding employers from firing workers for filing workers' compensation claims or for whistleblowing—led to a substantial reduction in injuries. The widespread adoption of the public policy exception explains 14% of the decline in fatal injury rates between 1979 and 1994. Statutory protections from retaliatory firing also improved safety, but only when employers faced sufficiently strong penalties for violating them.

I. Introduction

WORKPLACE injuries and illnesses impose a substantial economic burden; in the United States, their costs, both direct (for example, medical care) and indirect (for example, lost productivity), total roughly $250 billion each year (Leigh, 2011). Laws exist to ensure that employers internalize some of these costs. For example, the price of workers' compensation insurance, which firms are required to hold, is a function of their workers' claim history (Moore & Viscusi, 1989; Ruser, 1991). Firms are also required to comply with government safety regulations, which—to the extent that such regulations effectively address workplace hazards—ensure that firms face financial penalties for maintaining hazardous workplaces.

Historically, though, the doctrine of at-will employment has effectively allowed employers to treat employment relationships as outside the bounds of these laws. Under this doctrine, which has been the default law governing employment contracts since the late nineteenth century, an employer can dismiss a worker for any reason (that is, without having to establish "just cause"). Thus, in the absence of legal exceptions (which we discuss below), employers are permitted to discharge at-will employees in retaliation for filing workers' compensation claims or reporting violations of safety regulations to federal agencies—even though workers have a legal right to take these actions. Such retaliation is not uncommon: the Occupational Safety and Health Administration (OSHA) receives thousands of complaints about whistleblower retaliation each year (Weatherford, 2013). In the global COVID-19 pandemic of 2020, there have been numerous reports of employers firing workers in retaliation for speaking up about safety concerns.[1] If the threat of dismissal reduces workers' willingness to file for workers' compensation after injury, or to report unsafe or illegal activity, then employers may not internalize as many of the costs of workplace injuries as laws and policies intend.

In the 1970s, however, the relationship between at-will employment and workplace safety began to change in some states. The most salient reason for this change was an expansion of the common-law *public policy exception* to at-will employment, an exception which forbids employers from discharging employees who are either following or refusing to violate public policy.[2] First adopted in 1959 in California to protect employees refusing to commit perjury, the public policy exception expanded to protect dismissals for filing a workers' compensation claim in 1973 and whistleblowing in 1981. During this period many states also passed statutory protections (that is, laws passed by state legislatures) against dismissals for workers' compensation filing or whistleblowing.

We investigate how these legal protections against retaliatory discharge affected the likelihood that a worker would be injured on the job in the first place. It is plausible that such protections could improve workplace safety. If injured workers are more likely to file for workers' compensation, or if all workers are more likely to complain about unsafe conditions, then these legal protections raise employers' incentives to mitigate workplace injury hazards, leading to fewer future injuries.[3] At the same time, such protection could theoretically have the opposite effect and *increase* injuries: by lowering *workers'* effective cost of filing for workers' compensation, the exception could reduce workers' cost of getting injuried, incentivizing them to take less care on the job (Krueger, 1990).

To examine this relationship empirically, we leverage staggered adoption of the common-law public policy exception to at-will employment across states between the 1970s and 1990s. Additionally, we estimate the effects of *statutory* dismissal protections using newly quantified data on states' adoptions of statutory protections against dismissals for workers' compensation filing and whistleblowing. We

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[1]Examples include reports of Amazon firing warehouse workers (Day, 2020) and hospitals firing nurses (Carville et al., 2020) and doctors (Allen, 2020).

[2]The public policy exception is one of three exceptions to at-will employment adopted in the United States, alongside the implied contract and good faith exceptions. See section IIB.

[3]Exceptions to at-will employment could enhance workplace safety for other reasons. For example, safety is the result of relationship-specific investments, which are likely underprovided in at-will employment when turnover is inefficiently high (MacLeod & Nakavachara, 2007).

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examine the effects of these protections on rates of work-related injuries and illnesses with a difference-in-difference design, using multiple historical sources to measure injuries.

We find that common-law protections against retaliation for workers' compensation and whistleblowing led to a substantial improvement in workplace safety. States' adoption of the public policy exception led to an 11%–13% reduction in workplace injury and illness rates. This estimate is statistically significant and remains stable with the use of alternative specifications, additional control variables, and different data sources to measure injuries. Event study estimates reveal that this effect shows up immediately—the year following adoption—and persists for several years.

To put this magnitude in context, fatal workplace injury rates in the United States declined 41% between 1980 and 1994. Considering the change in the share of the workforce that was covered by the public policy exception over this same period, our estimates imply that the widespread adoption of the public policy exception explains roughly 14% of this overall decline.

In contrast to the common law public policy exception, we find less consistent evidence that statutory protections improved safety. We find suggestive evidence that statutory protection for whistleblowing improved safety: the point estimates indicate a negative effect on injuries, but the magnitude and statistical significance are somewhat sensitive to specification. We find no evidence that the workers' compensation statutory protection had any safety effect: our point estimates are all essentially zero and never statistically significant. Given the institutional details of these statutory protections, these more muted average effects of statutory protections are not necessarily surprising: the typical statutory protection yielded lower expected costs to employers than did the public policy exception to at-will employment, and their scale and stringency varied widely across states. Heterogeneity analysis supports this interpretation: while statutory protections had a small effect on average, they led to a larger reduction in injuries in states in which the protections levied larger penalties on employers that violated them.

These results suggest that the public policy exception to at-will employment incentivized employers to take costly actions to reduce workplace injuries and illnesses. However, other mechanisms are also possible. We conduct two analyses to assess whether ours is the likely mechanism at play. First, we find that the public policy exception led to improved compliance with safety and health regulations–a salient example of employers' inputs into abating workplace hazards. Second, we assess whether the effects of the exception are heterogeneous based on factors that we expect would affect the degree to which the public policy exception raised employers' costs of workplace injuries.

To this end, we assess whether the safety effect of the public policy exception depends on the presence of labor unions. As we discuss in section VIC, much evidence indicates that unions lower the cost to individual workers of filing for workers' compensation, such as by collecting information about relevant laws and policies (Weil, 1996) as well as providing direct support for the filing of claims (Hirsch et al., 1997). Unions similarly facilitate workers' blowing the whistle. Thus, to the extent that the public policy exception raised the threat that a worker would file for workers' compensation following an injury or blow the whistle about unsafe conditions, this threat would have been more credible among unionized workers.[4]

Consistent with this logic, we find that the public policy exception led to a substantially larger reduction in injuries among unionized employers than among nonunionized employers. Bolstering this evidence of the role of unions, we find that the public policy exception had a substantially smaller effect on injuries in states that had adopted right-to-work (RTW) laws—which have been shown to substantially weaken the power of unions—as of the start of our sample period.

Our results extend a robust literature on the effects of exceptions to at-will employment (or wrongful discharge laws) on labor markets. Prior studies have examined the effects of these exceptions on employment and wages (Autor et al., 2006), productivity (Autor et al., 2007), innovation (Acharya et al., 2013), and investments in relationship-specific assets (MacLeod & Nakavachara, 2007), among other outcomes. We contribute to this literature by investigating a previously unexplored effect of these exceptions and revealing two sources of heterogeneity in their effects.

Our paper also adds to a wide literature on the economic determinants of workplace safety and health (surveyed in Ruser & Butler, 2010), and in particular on the effects of workers' compensation (surveyed in Kniesner & Leeth, 2014). Prior work has examined the effects of insurance premiums (Moore & Viscusi, 1989), experience rating (Ruser, 1991; Thomason & Pozzebon, 2002), and benefit levels (Fishback, 1987; Krueger, 1990). We show that common-law protections making it less risky for workers to blow the whistle or file for workers' compensation improved workplace safety—with premiums, benefit levels, and other factors held constant.

II. Background

Even though employers have some private incentives to limit workplace injuries, the institution of at-will employment, policy distortions, and labor market frictions likely attenuate such incentives. As a result, legal restrictions on retaliatory firing—which have been adopted in various forms across a subset of states—could motivate employers to invest more in worker safety, thus reducing the occurrence of work-related injuries.[5]

[4]It is also possible that the public policy exception would be redundant in unionized firms since unions already include "just cause" provisions in contract negotiations. We discuss this alternative possibility in section VIC.

[5]For more details surrounding the discussion we present in this section, see appendix A.

A. *Workers' Compensation, Safety Regulations, and Employers' Costs of Injuries*

Existing public policies and labor market competition, in theory, ensure that employers face incentives to mitigate workplace injuries. However, much evidence suggests that these disciplinary forces are more muted than might be expected.

One such policy is the workers' compensation system. Passed in the United States in the early 1900s, this system insures workers against income risks in the event of a job-related injury. The premiums that most employers pay into the workers' compensation system are "experience-rated," meaning that they depend on the employer's history of prior claims. Furthermore, employers pay deductibles under most plans, ensuring that they pay at least a portion of total injury costs. Because these features raise employers' costs when they experience more injuries, they provide incentive for employers to improve worker safety.[6]

Other public policies enable workers to file safety and health complaints with the federal government and to serve as whistleblowers. Section 11(c) of the Occupational Safety and Health Act of 1970 gives workers the right to file complaints with the Occupational Safety and Health Administration (OSHA) when they feel exposed to a serious hazard or that their employer is violating safety and health regulations. OSHA, the federal agency charged with ensuring "safe and healthful working conditions" (OSHA, 2020), also enforces various whistleblower laws that, in principle, protect workers from discharge for blowing the whistle in domains ranging from asbestos removal to consumer protection. In theory, workers' ability to complain or blow the whistle motivates companies to proactively improve safety (Weil & Pyles, 2005).

However, evidence suggests that employers do not fully internalize the costs of injuries the way that these policies intend. First, up to half of eligible injuries do not get filed for workers' compensation (Shannon & Lowe, 2002; Biddle & Roberts, 2003; Fan et al., 2006; Groenewold & Baron, 2013). One reason that would certainly deter injured workers from filing for workers' compensation is if they perceive a threat of retaliation from their employer for doing so (Spieler & Burton Jr., 2012). One recent study found that 20% of workers reported fearing that they could lose their job if they filed a workers' compensation claim (Edisis, 2017). Another found that 50% of low-wage workers in New York City, Los Angeles, and Chicago reported being instructed not to file a workers' compensation claim or were fired for doing so (Bernhardt et al., 2009).

Similar barriers limit workers' ability to take advantage of their rights to complain or to blow the whistle. Punishment for violating Section 11(c) is essentially nonexistent: employers face no fines if they violate it. Furthermore, a high burden of proof and other barriers make it difficult for workers to file a whistleblower complaint to OSHA (Weatherford, 2013). A 1990 report found that fewer than 10% of OSHA's own inspectors said that workers could definitely exercise their rights to complain without fear of retaliation (Government Accountability Office, 1990).

How would these policy distortions affect the provision of workplace safety? Injuries are generally more costly for employers when the injured worker files for workers' compensation, since most claims raise the employer's future premium and entails paying a deductible.[7] Similarly, maintaining a hazardous workplace is more costly for employers if their workers complain or blow the whistle to OSHA, since these actions can lead to regulatory fines and bad publicity (Johnson, 2020). If contracts are incomplete, an employer has the incentive and ability to fire an at-will worker in retaliation for filing a workers' compensation claim or for blowing the whistle. Workers, fearing a threat of dismissal, will be less likely to undertake these actions. Injuries are thus less costly for employers in expectation, reducing employers' incentives to make investments to reduce them. As a result, laws that limit employers' ability to retaliate against workers for filing workers' compensation claims or blowing the whistle could raise employers' investments in worker safety.[8]

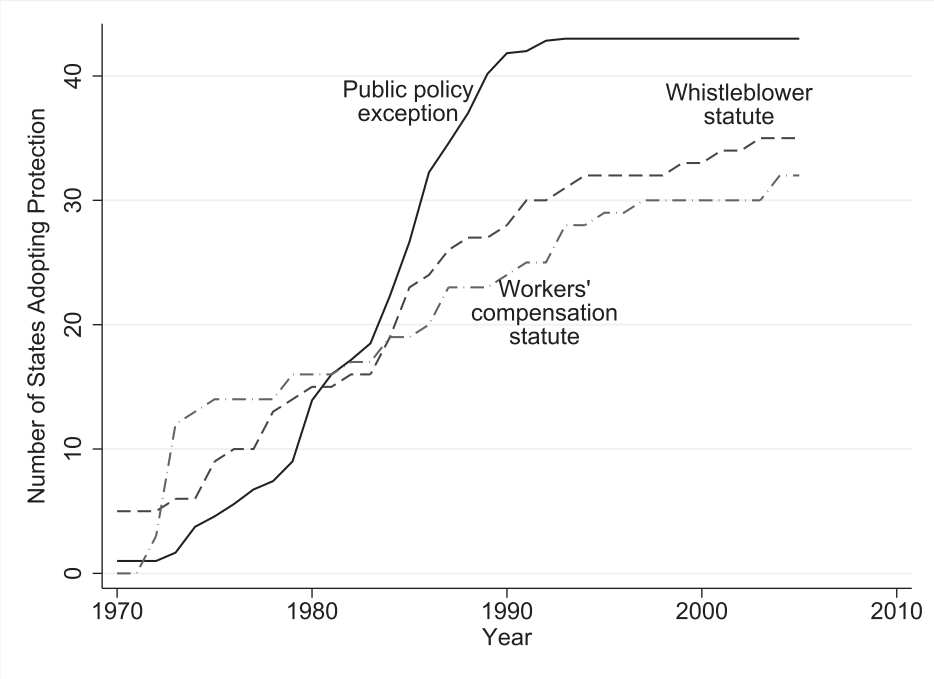
However, even if such policy distortions are present, features of the labor market presumably already incentivize employers to limit injuries. In a competitive labor market, workers demand higher wages to work in riskier jobs (Rosen, 1986); indeed, much empirical evidence in the compensating differential literature confirms that workers earn higher wages for undertaking riskier jobs (Viscusi & Aldy, 2003; Kniesner et al., 2012; Lee & Taylor, 2019; Lavetti, 2020). However, growing evidence that most labor markets are characterized by imperfect competition suggests that this market discipline might be more muted than implied in canonical models. Monopsonistic competition is pervasive (Manning, 2011; Dube et al., 2020), arising from explicit sources like employer concentration (Azar et al., 2020) but also broader factors like idiosyncratic worker preferences (Lamadon et al., 2022). Imperfect competition affects both the *level* of (wage and nonwage) compensation (Prager & Schmitt, 2021; Dube et al., 2018), but it also attenuates the *price* of injury risk with respect to wages (Lavetti & Schmutte, 2018). Workers also have imperfect information

[6]Prior studies have demonstrated, as theory would predict, that experience rating (Ruser, 1991; Bruce & Atkins, 1993; Thomason & Pozzebon, 2002), higher premiums (Moore & Viscusi, 1989), and higher deductibles (Shields et al., 1999) all lead to fewer workplace injuries. See Kniesner and Leeth (2014) for an overview of the literature examining the incentive effects of different aspects of the workers' compensation system.

[7]One exception is that particularly small firms pay a flat workers' compensation that is not experience rated, but such firms employ a small share of the overall workforce (Ruser, 1985).

[8]Acharya et al. (2013) show theoretically that incomplete contracts create a similar hold-up problem for employee innovation effort: employers cannot commit to not armtwist employees who contributed considerable effort to valuable innovation for a larger share of the ex-post surplus. As a result, innovation effort is inefficiently low in at-will employment relationships, and laws that limit employers' ability to engage in such retaliation raise employees' innovative effort.

FIGURE 1.—STATES' ADOPTION OF PUBLIC POLICY EXCEPTION AND WORKERS' COMPENSATION AND WHISTLEBLOWER STATUTES OVER TIME



This figure shows the number of states adopting the public policy exception, workers' compensation statutes, and whistleblower statutes over time. Please see text for sources.

about injury risk and other job attributes (Viscusi & Moore, 1991; Conlon et al., 2018). Thus, while the labor market undoubtedly ensures that employers have some private incentive to minimize injury risk, imperfect competition attenuates this incentive.

Finally, even if one abstracts from policy distortions and imperfect competition, workplace safety still might be inefficiently low under at-will employment. Investments in workplace safety include capital expenditures like updating machinery, but they also include relationship-specific assets like worker training and developing familiarity with processes, equipment, and "culture" (Williamson et al., 1975). If contracts are incomplete and cannot be conditioned on the level of relationship-specific investment, then parties will under-invest in these relationships under at-will employment (MacLeod & Nakavachara, 2007).

Policy distortions likely mute the extent to which workers' compensation and whistleblower opportunities incentivize employers to improve safety in at-will employment relationships. Furthermore, imperfect competition and incomplete contracts attenuate the extent to which the labor market disciplines employers' provision of safety. Legal protections against employer retaliation, by raising employers' expected costs of injuries and by incentivizing investment in relationship-specific assets, could thus improve workplace safety.

B. *Legal Restrictions on Retaliatory Firing*

Over the last few decades, various states have adopted two types of limits on employers' ability to retaliate against workers for filing for workers' compensation or blowing the whistle.

The public policy exception to at-will employment. Common law is adopted through precedent that arises from the decisions of particular court cases. Since the late nineteenth century, U.S. courts have generally interpreted the employer-employee relationship to be one of equal power for both parties; the resulting "at-will" doctrine concluded that any employment contract could be terminated at any time by either party. However, beginning with the Industrial Revolution, judges began to recognize "wrongful discharge laws," or exceptions to this interpretation that reflected recognition of power disparities between employers and employees.

The at-will exception most relevant to the current paper is the *public policy exception.* First adopted in California in 1959, the public policy exception prohibits the dismissal of an employee who is either following or refusing to violate well-established public policy. The exception initially protected a relatively narrow set of actions, including serving on a jury or refusing to commit perjury. Legal cases in 1973 and 1981 expanded the exception to protect workers against retaliation for filing workers' compensation claims and for whistleblowing, respectively. In 1970, California was the sole adopter of the public policy exception. By 1980, 15 states had adopted the exception, and this number grew to 42 by 1990. Figure 1 shows the number of states that had adopted the exception each year since 1970.

Once adopted, the public policy exception represented a salient change in the legal environment for employers. It is

a tort-based action, meaning employees can sue for not just compensatory damages (for example, back pay, attorney's fees), but also punitive damages. Because punitive damages are meant to "punish" the employer, they can be considerable (Edelman et al., 1992).

Along with the public policy exception are two other recognized common-law exceptions to at-will employment, which are not the focus of our paper. The *implied contract exception* prevents dismissals that are in violation of a written or verbal statement that implies a contract has been established. The *good faith exception* establishes a covenant of good faith and fair dealing in all employer-employee relationships, effectively requiring all dismissals be made with just cause, though in practice it is mainly applied in cases related to the timing of a dismissal.

Prior studies indicate that passage of the public policy exception was not driven by underlying political or economic trends (Autor et al., 2007; DeNicco, 2015). Instead, common-law exceptions are a function of cases specific to a particular employment relationship or occupation and the willingness of sitting judges to hear them. Bird and Smythe (2008) find that neither economic nor political factors had any significant predictive power for if and when a state adopted the public policy exception; the authors conclude that "judges usually base their decisions on legal authorities rather than policy considerations or economic conditions" (Bird & Smythe, 2008).

Statutory workplace safety protections. Whereas common law is adopted through court precedent, states' statutory law is encoded into legislation passed by state legislatures. Many states have adopted statutory protections for filing for workers' compensation and for whistleblowing. While these protections protect a similar scope of worker actions as the common-law public policy exception, legal scholars have argued that most statutory protections are less likely to be an effective deterrent for employers (Sinzdak, 2008).

Thirty-five states have enacted whistleblower statutes that forbid employers from terminating employees for reporting unlawful activity within the firm, such as noncompliance with safety and health regulations. However, most statutes impose burdens that make it difficult for individuals to initiate a claim (see appendix A for details). Furthermore, the damages that claimants can pursue are heterogeneous across states: in most states claimants are limited to pursuing compensatory damages, and only a few allow for punitive damages. These limitations are in contrast to the pubic policy exception, which offers a clear, uniform course of action for employees to file suits for both compensatory and punitive damages (Edelman et al., 1992).

Additionally, 35 states have enacted statutes to prohibit dismissal in response to an employee's filing for workers' compensation. Punishments prescribed by these statutes vary widely, but they tend to be even lower than those under the whistleblower statutes (see appendix A for details).

TABLE 1.—SUMMARY STATISTICS

	Mean	SD
Injury inspections per 1000 employees (OSHA)	0.031	0.059
Deaths per 1000 employees (NSC)	0.061	0.046
Public policy exception	0.548	0.494
Workers' comp. antiretaliation statute	0.449	0.497
Whistleblower protection statute	0.419	0.494

Table shows summary statistics for our key dependent and explanatory variables. Please see text for further details and sources.

We describe each state's whistleblower and workers' compensation statutes in tables B1 and B2. Figure 1 shows how many states had adopted these statutes each year.

Given the relatively low and variable levels of potential damages available to claimants under statutory protections for whistleblowing and filing workers' compensation, we expect them to have (a) less of an average effect on injuries than the common-law public policy exception, which poses a stronger and more consistent means of preventing retaliatory discharge, and (b) a larger effect in states that enable claimants to pursue punitive damages.

III. Data

To undertake our analysis we need data on (1) the years in which states adopted protections against retaliation and (2) measures of workplace injuries. We present summary statistics for these measures in table 1 and describe them in this section.

A. *Adoption of the Public Policy Exception and Statutory Protections*

To measure adoption of the public policy exception, we use data from Autor et al. (2006) on precedent-setting legal decisions that signify the adoption years of each of three exceptions to at-will employment. We create a variable equal to 0 if a state has not adopted the exception, to 1 in the years the state adopted the exception, and to a fraction between 0 and 1 in the year of adoption depending on the month it occurred. We create analogous variables for the good faith and implied contract exceptions to at-will employment, which serve as controls.

We use a combination of sources to measure the adoption of statutory protections for workers' compensation filing and whistleblowing. We obtained workers' compensation statutory codes from Littler's Workers' Compensation Retaliation Survey (Altman et al., 2012), which we cross-referenced with state law codes available through LexisNexis as well as additional law review journals to confirm passage years. We conducted similar cross-referencing searches for whistleblower statutes using statute codes and passage years provided by Callahan and Dworkin (2000) and statutes collected by The Employment Law Group P.C. and the National Conference of State Legislatures. Using these data, we construct a panel from 1970 to 2005 of state-year observations

for adoption of both the workers' compensation and whistleblower statutory protections.[9]

B. *Measuring Workplace Injuries and Illnesses*

We use two distinct sources to measure work-related injuries and illnesses over the period in which legal protections against retaliatory dismissal were enacted.

Our first measure is the occurrence of OSHA inspections triggered by a serious workplace injury. Many employers are required to comply with hundreds of OSHA regulatory standards, which range from maintenance of specific capital equipment to more general restrictions that workers not be exposed to certain hazards. OSHA has direct jurisdiction in 29 states; the remaining 21 states have received approval to operate their own state-run safety and health programs.[10] Inspections are OSHA's primary tool for monitoring compliance with these standards. OSHA inspections can be initiated for multiple reasons. Most relevant to our paper, in the event that a workplace experiences a worker fatality or hospitalization of three or more workers, the employer is required by law to report it to OSHA, and OSHA is required to inspect the workplace. Thus the occurrence of such an inspection indicates that a serious injury took place.

The majority of OSHA's inspections occur for reasons other than a serious injury. "Programmed" inspections, which focus on particular industries or hazards, are initiated for reasons exogenous to events at a particular workplace.[11] These inspections are pursuant to National Emphasis Programs (NEPs), which focus on nationwide priorities, or Local Emphasis Programs (LEPs), which focus on regional priorities. Because programmed inspections target workplaces only in a particular industry or that are likely to have a specific hazard (for example, based on their location or production technology), no workplace-specific factors (such as recent injuries) influence the occurrence of a programmed inspection.[12] Furthermore, conditional on the criteria on which NEPs or LEPs are based (for example, industry or region), OSHA often allocated inspections among establishments meeting these criteria using random assignment (Lee & Taylor, 2019).

We identify the occurrence of OSHA inspections (and those conducted by its state counterparts) using OSHA's Integrated Management Information System (IMIS), a database that contains detailed information on every OSHA inspection.[13] Key variables include the date the inspection was opened, the reason for the inspection (injury, complaint, referral, programmed, other), and facility characteristics (such as name, address, industry, number of employees, and whether the employees were represented by a labor union.).

While states under OSHA's jurisdiction began reporting to the IMIS database in 1979, the 21 states outside of OSHA's jurisdiction did not begin reporting to IMIS until the late 1980s, with complete records starting in 1992. Thus we restrict our analysis of the data to inspections that occurred beginning in 1979 in states under federal OSHA jurisdiction and inspections that occurred beginning in 1992 in states outside federal OSHA jurisdiction. We collapse the inspection data to the state-year level to obtain the number of inspections in each category occurring each year in every state (in robustness checks, we collapse the data at the more refined state-sector-year level).

As a secondary measure of workplace injuries and illnesses, we digitized historical records of annual work-related fatalities from the National Safety Council (NSC) in its annual publication *Accident Facts* for the years 1970 through 2000. These reports included tables on the principal classes of accidental deaths by state. Most relevant to this paper, these classes include fatal workplace injuries, which the NSC collected from state industrial commissions. States reported these data to NSC on a voluntary basis; as a result, some states did not report in certain years.

Each of our two measures of workplace injuries has relative advantages and disadvantages. One advantage of the OSHA data is that it captures both fatal and serious nonfatal injuries, thus covering a larger share of work-related injuries than the NSC data. Further, at least for the states under federal OSHA jurisdiction, the OSHA data provide a balanced panel of injury rates over several decades. On the other hand, the NSC data are based on actual injury reports (rather than on government inspections resulting from an injury), the reporting begins in 1970 (nine years before any OSHA data), and fatal injuries are reported with high accuracy (Morantz, 2013). However, because states voluntarily reported their data to the NSC, data availability is intermittent.

With these comparisons in mind, a benefit of having both the OSHA and NSC measures is that we can assess the effects of legal protections on safety using two independent measures of injuries that cover a different distribution of states over time. Figure D2 plots the annual coverage of the NSC and OSHA data from 1970 through 2005.

C. *Constructing Injury and Illness Rates*

To measure work-related injury and illness *rates*, we obtain total employment in each state-year from Current

[9]Unlike the public policy exception, we do not observe the month of adoption for statutory protections. We code these variables as indicators equal to one starting the year of adoption.

[10]Figure D1 shows which states are under OSHA's jurisdiction.

[11]Inspections can also be triggered by a complaint alleging safety and health hazards, or a "referral" (an allegation of hazards made by an inspector, a government agency, or the media).

[12]The exception to this statement is OSHA's Site-Specific Targeting program, formed in 1998 to focus a subset of programmed inspections on establishments that had recently experienced high injury rates. However, this program is not highly relevant to our study, as the bulk of our sample period is prior to 1998.

[13]We downloaded the data from OSHA's website in July 2014, available at https://enforcedata.dol.gov/views/data_summary.php.

Employment Statistics, published by the Bureau of Labor Statistics. In some analyses, we measure injury rates separately for unionized and nonunionized workers using data on the share of private-sector workers that are unionized by state-year from the website unionstats.com, with data based on Hirsch and Macpherson (2003).[14] We multiply the total employment by the percentage unionized to obtain the number of unionized and nonunionized workers in each state-year.

IV. Empirical Strategy

Our goal is to test how legal restrictions against an employer's ability to retaliate against a worker for filing a workers' compensation claim or for reporting illegal conditions to the government affect workplace safety. We focus on the common-law public policy exception as our primary measure of such legal protection in developing our empirical strategy, but we use the same strategy to examine the effects of statutory protections against retaliation.

To identify our intended effect, we must address several sources of endogeneity. First, states that adopt the public policy exception (or statutory protections) might have time-invariant characteristics that lead them to experience more or fewer injuries than nonadopting states for unrelated reasons (for example, states that adopt the exception might tend to have an overall legal environment more favorable to workers). Second, trends in adoption of the public policy exception and in workplace safety changed over time. We include state and year fixed effects to account for these concerns.

There may still be other differences between adopting and nonadopting states. Prior studies have shown that adoption of common-law exceptions has strong regional variation; for example, Southern states lagged behind other regions in adopting the good faith exception (Autor et al., 2006), and injury rates may vary across regions for other reasons, such as differences in industry mix, worker characteristics, or temperature. We follow prior studies in this literature and include fixed effects for each year interacted with the nine U.S. Census divisions. Further, injury rates in states that adopted the public policy exception could be on a different trajectory than those in states never adopting it.[15] We address this final concern by including state-specific linear time trends in some specifications.

In light of the preceding considerations, we follow several prior studies in the literature (for example, Autor et al., 2006, 2007; MacLeod & Nakavachara, 2007; Acharya et al., 2013) to leverage the staggered adoption of the public policy exception across states to estimate its effect on safety with a difference-in-difference design. Our main specification is as follows:

$$\ln y_{st} = \beta \cdot PPE_{st} + \delta_s + \varphi_s I_s \cdot t + \tau_{dt} + X_{st}\gamma + \epsilon_{st}, \quad (1)$$

where y_{st} is the dependent variable observed for state s in year t. Our primary dependent variable is the log injury rate per 1,000 workers. In our main specifications, we measure the injury rate per 1,000 workers as the number of OSHA inspections triggered by a serious workplace injury divided by total state employment (in thousands). In an alternative specification, we use the number of workplace deaths recorded by the NSC in place of OSHA injury inspections.

The explanatory variable of interest, PPE_{st}, is the indicator equal to 1 if a state has adopted the public policy exception to at-will employment. The term δ_s is a state fixed effect, τ_{dt} represents a division-year fixed effect, and $I_s \cdot t$ represents state-specific linear time trends (with I_s representing state dummies). In some specifications, we include the following additional control variables in X_{st}: indicators that a state has adopted the good faith and implied contract exceptions to at-will employment, the state unemployment rate (because workplace accidents might be influenced by macroeconomic conditions (Boone et al., 2011)), the political party of the governor (as a proxy for the political climate, which could affect workplace safety), and the log of the number of OSHA programmed inspections per thousand workers in the state-year (to account for potential variation in OSHA's overall inspection intensity). In all regressions, we cluster standard errors by state to allow for heteroskedasticity and arbitrary correlation of the error term within a state.

As described above, the years that we include in our sample depend on the dataset used. When the dependent variable is measured using OSHA inspections, we include data beginning in 1979 for states without state-run OSHA offices and beginning in 1992 for states with state-run OSHA offices. When we use NSC data to measure the dependent variable, we begin the sample period in 1970. We end our sample period for all states in 2005; we choose this end date because the latest adoption of the public policy exception occurred in 1990, and the latest statutory protection was adopted in 2003 (we consider alternative end years in robustness checks). We weight observations by each state's share of the national labor force in 1979 when measuring injuries with OSHA data, and by the share in 1970 when measuring injuries with NSC data.

Our identifying assumption is $E(\epsilon_{it} PP_{st} | \delta_s, I_s \cdot t, \tau_{dt}, X_{st}\gamma) = 0$; that is, conditional on state fixed effects, region-year fixed effects, state-specific trends, and additional controls, adoption of the public policy exception (or statutory protections) is exogenous to injuries. This assumption would

[14] unionstats.com reports data on overall state-level unionization rates starting in 1964, but the earliest year it reports data on unionization rates separately for the private and public sector is 1983. Because our OSHA data begin in 1979, we impute states' private-sector unionization rates for the years 1979–1982. For each state, we compute the percent difference between the private sector and overall unionization rate in each year 1983–2016. We then regress this percent difference on a state-specific time trend. We use the fitted line to predict the 1979–1982 ratio of private to overall unionization. Finally, we multiply this predicted ratio by the overall unionization rate to get our estimate of each state's private sector unionization rate for the years 1979–1982.

[15] This could be the case, for example, if states with especially fast declines in injury rates are less likely to have employment disputes leading to a court decision on the public policy exception.

TABLE 2.—THE EFFECT OF THE PUBLIC POLICY EXCEPTION AND STATUTORY PROTECTIONS ON WORKPLACE INJURIES (DATA FROM OSHA)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Public policy exception	−0.137***			−0.123***	−0.120***	−0.095**	−0.112*
	(0.046)			(0.046)	(0.044)	(0.046)	(0.062)
Whistleblower statute		−0.087**		−0.072*	−0.072**	−0.056	−0.114*
		(0.037)		(0.039)	(0.034)	(0.053)	(0.062)
Workers' comp statute			−0.032	−0.003	0.027	0.077	0.014
			(0.044)	(0.034)	(0.025)	(0.056)	(0.086)
Good faith exception					−0.203***	−0.176**	−0.095
					(0.041)	(0.077)	(0.136)
Implied contract exception					−0.024	−0.029	−0.003
					(0.047)	(0.062)	(0.091)
State unemployment rate					−0.034***	−0.038***	−0.023*
					(0.012)	(0.011)	(0.014)
Democratic governor					−0.024	−0.037	−0.024
					(0.039)	(0.028)	(0.026)
Prog. inspection rate					0.020	0.054	−0.013
					(0.038)	(0.045)	(0.037)
Observations	1,077	1,077	1,077	1,077	1,077	1,077	1,077
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State trend	No	No	No	No	No	No	Yes
Division-year FE	No	No	No	No	No	Yes	Yes

The dependent variable is the natural log of the number of OSHA injury inspections divided by the state labor force (in thousands). The primary independent variables are dummy variables for whether a state has adopted the public policy exception or whistleblower or workers' compensation statutes. Regressions are weighted by state employment in the first year of the sample, with robust standard errors clustered by state in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

be violated if, for example, adoption of legal restrictions on at-will employment is correlated with states' underlying (unobserved) economic or political trends. However, given the institutional background described earlier, this is unlikely a concern in practice. Because judges cannot make precedent-setting decisions unless appropriate cases are available, the adoption of a particular exception is difficult to predict and can be due to largely idiosyncratic factors, limiting potential behavior that could introduce endogeneity. We note endogeneity is potentially a more salient concern when considering adoption of statutory protections, but evidence we present later suggests this is unimportant in practice.

Finally, to ensure that our estimates are not dependent on any arbitrary choices in our specification, we report several variations on equation (1), such as regressions that omit division-year fixed effects and/or omit state-specific linear trends.

V. The Safety Effects of Legal Protections against Retaliatory Firing

We present our estimates of the effect of the public policy exception to at-will employment and the two statutory protections on workplace injury rates in section VA. We assess the sensitivity of these estimates to alternative modeling choices and data sources, and to potential bias arising from heterogeneous treatment effects, in section VB.

A. *Baseline Estimates*

Table 2 presents our main estimates of the effects of various legal protections against retaliatory firing on workplace injury and illness rates, measuring injuries and illnesses with the OSHA data. Columns 1–3 reports estimates from a regression with state and year fixed effects only, for our three forms of protection. The coefficient reported in column 1 indicates that adoption of the public policy exception to at-will employment led to 12.8% fewer injuries per worker ($\hat{\beta} = -0.137$, $\exp(-0.137) - 1 = -0.128$), and the estimate is highly significant ($p = 0.006$). Adoption of the statutory protection for whistleblowing (column 2) led to an estimated 8.3% fewer injuries ($\hat{\beta} = -0.087$, $p = 0.023$). Adoption of the statutory protection for filing workers' compensation (column 3) had no detectable effect on safety ($\hat{\beta} = -0.032$, $p = 0.47$).

In the remaining columns, we successively consider various concerns with the specification in columns 1–3. In column 4, we include all three protections in the same regression. In column 5, we additionally control for states' adoption of the other two common-law exceptions to at-will employment—good faith and implied contract—as well as the three additional controls described above. In column 6, we replace year with division-year fixed effects, and in column 7, we include state-specific linear trends. The coefficient on the public policy exception remains statistically significant at least at the ten percent level, and the magnitude is stable across these specifications.

The coefficient on the whistleblower statutory protection attenuates slightly in magnitude to −0.072 in columns 4 and 5, loses statistical significance in column 6 with division-year fixed effects, but then slightly *increases* in magnitude in column 7 with the inclusion of state-specific trends ($\hat{\beta} = -0.114$, $p = 0.076$). The coefficient on the workers' compensation statute remains tiny and statistically insignificant across all columns.[16]

[16]While not the focus of our study, the coefficients on the other two wrongful discharge laws are of interest. The coefficient for the good faith

The difference-in-difference specification in equation (1) imposes the assumption that the effect of legal protection against retaliatory firing on workplace injuries is constant over time. Furthermore, the estimates will be misleading if states that adopted legal protections were experiencing differential trends in injuries in the years prior to adoption.

To assess these concerns, we consider the dynamic impacts of the adoption of the three legal protections against retaliation using the following event study specification:

$$\ln y_{st} = \sum_{i=-5}^{5} \beta_i \cdot PPE_{st}^{i} + \beta_6 PPE_{st}^{6+} + \beta_{-6} PPE_{st}^{-6+} + \delta_s + \varphi_s I_s \cdot t + \tau_{dt} + X_{dt} + \epsilon_{st}, \quad (2)$$

where PPE_{st}^{i} is equal to 1 if the state adopted the public policy exception exactly i years ago, PPE_{st}^{6+} equals 1 if the public policy exception was adopted six or more years in the past, and PPE_{st}^{-6+} equals 1 if the public policy exception was adopted six or more years in the future. The rest of the regression is the same as the main regression (equation (1)), with state-year controls, state-specific linear trends, and division-year fixed effects included. We estimate analogous models for adoption of the two statutory protections.

Figure 2a plots the coefficients on the leads and lags of adoption of the public policy exception. We normalize the coefficient on PPE_{st}^{-1} to be zero. The coefficients on the lead terms are all close to zero, indicating that no important pretrends in injury rates existed in the years prior to adoption of the public policy exception. The lag coefficients indicate a moderate decline in injuries in the year that the exception was adopted, and then a larger decline in the year following passage and persisting for at least five years.

We report analogous event study results for the whistleblower statutory protection in figure 2b. The coefficients on the lead terms are all close to zero; after the passage of a whistleblower protection statute, the coefficients become negative, albeit most terms individually are not significantly different from zero. Figure D3 presents the event study for the workers' compensation statute: corroborating the regression results, there is no evidence of a change in injuries following its adoption.

B. Robustness Tests on Baseline Estimates

Our estimates are stable to alternative datasets and specification choices, are corroborated by placebo and falsification tests, and are robust to potential bias from heterogeneous treatment effects.

Alternative data source to measure injuries. The prior section revealed that the estimated effect of the public policy exception on workplace injuries is highly robust to different choices of fixed effects, state trends, and inclusion of control variables when we measure injury rates using the OSHA data. Our estimates are also insensitive to using an alternative measure of workplace injuries. Rather than use the number of OSHA inspections triggered by a serious accident to measure the number of injuries in a state-year, we use the number of fatal workplace injuries from the NSC (described above). As with the OSHA measure, our dependent variable is the natural log of the rate of fatal workplace injuries per 1,000 workers. Our specification is otherwise the same as equation (1), except that we weight observations by each state's 1970 (rather than 1979) employment since the NSC data begins in 1970.

Table 3 presents estimates from nearly identical regressions to those in table 2, but the dependent variable is now NSC's fatal injury rate rather than OSHA's injury inspection rate. In column 1, the simplest specification with state and year fixed effects and no other controls, the coefficient on *Public Policy Exception* is -0.133 ($p < .01$), which is essentially identical to the analogous estimate with the OSHA data in table 2 (-0.137); this similarity is all the more remarkable given that the time period and set of states are quite different in these two tables. Across the remaining columns, the coefficient on the public policy exception remains stable and similar to estimates obtained with the OSHA data.[17] Across all specifications, the coefficients on the *Whistleblower statute* are negative, but they are smaller in magnitude than our estimates with the OSHA data and never statistically significant. As with the OSHA data, the coefficients on the *Workers' Compensation statute* are uniformly small in magnitude and nowhere near significant.

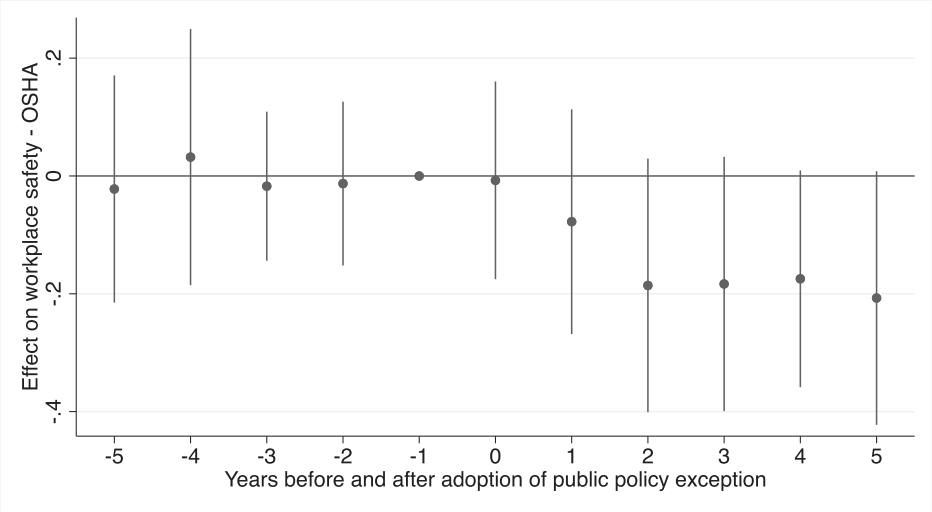
Appendix C.1 presents event studies for the safety effect of all three policies with NSC data, analogous to those we describe above using the OSHA data. These event studies are noiser than those using the OSHA data, but they broadly support the regression results reported in table 3.

Alternative specification choices. In appendix C.2, we show that our results are robust to sensible changes in the sample including: ending our sample period in earlier years, estimating a "donut regression" (excluding the years immediately before, after, and during a change in public policy exception adoption), and restricting the sample to those states under federal OSHA's purview. Our results also hold up against other specification choices, including separating the sample into manufacturing and nonmanufacturing, using alternate transformations of the dependent variable, and estimating Poisson or negative binomial regressions rather than our linear model.

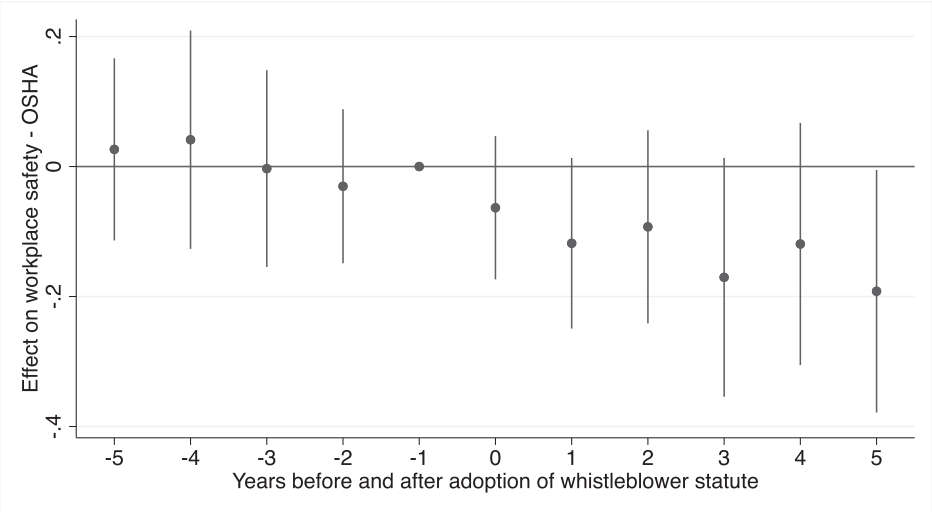
exception is large in magnitude and strongly statistically significant in the baseline specification with state and year fixed effects in column 5, but it attenuates by 50% and loses statistical significance with division-year and state trends in column 7. The point estimate for the implied contract exception is small in magnitude and not statistically significant.

[17] We exclude state unemployment as a control because the data begin only in 1976 (whereas the NSC data begin in 1970).

FIGURE 2.—EVENT STUDIES: EFFECTS OF THE PUBLIC POLICY EXCEPTION AND WHISTLEBLOWER STATUTE ON WORKPLACE INJURIES (DATA FROM OSHA)



(a) Effect of the Public Policy Exception on Workplace Injuries



(b) Effect of the Whistleblower Statute on Workplace Injuries

The plot shows the effect of the public policy exception and whistleblower statute on workplace injuries (data from OSHA) before and after adoption, based on equation (2). Coefficients on the left side of the plot indicate leads and coefficients on the right side indicate lags. Included in the regression but not displayed are coefficient terms for having adopted the relevant policy six or more years in the past or six or more years in the future; see text for details. The dots represent the point estimates, and the vertical bars show 95% confidence intervals.

Falsification tests. We present several falsification tests, which are discussed fully in appendix C.2.1. First, we test for the possibility that the public policy exception itself impacts the rate of OSHA inspections unrelated to injuries. Second, we examine whether there is a relationship between the public policy exception and non-work-related fatal injuries. Evidence of a relationship in either case would suggest that our main findings may be spurious, but we find no such evidence.

Alternative inference using placebo adoption. Appendix C.3 presents a complementary statistical approach to our main analysis where we generate alternative p-values using "randomization inference" (Young, 2019). The implied p-value is quite similar to the p-value we obtain from our main specification using standard inference techniques.

Assessing potential bias from heterogeneous treatment effects. Recent studies have illustrated that the standard "two-way fixed effects" estimator (which we employ) can yield a biased estimate of the average effect of a policy in the presence of heterogeneous treatment effects (de Chaisemartin & d'Haultfoeuille, 2020; Goodman-Bacon, 2021; Callaway & Sant'Anna, 2021). We probe the sensitivity of our estimated effects of the public policy exception on safety with an alternative approach developed in a recent paper (Callaway & Sant'Anna, 2021). In appendix C.4, we describe this approach in more detail, how we implement it,

TABLE 3.—THE EFFECT OF THE PUBLIC POLICY EXCEPTION AND STATUTORY PROTECTIONS ON FATAL WORKPLACE INJURIES (DATA FROM NSC)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Public policy exception	−0.133***			−0.143***	−0.132**	−0.108**	−0.109*
	(0.047)			(0.046)	(0.057)	(0.046)	(0.055)
Whistleblower statute		−0.029		−0.057	−0.057	−0.030	−0.061
		(0.062)		(0.061)	(0.053)	(0.044)	(0.049)
Workers' comp statute			−0.040	−0.039	−0.018	0.015	0.117
			(0.071)	(0.060)	(0.056)	(0.060)	(0.079)
Good faith exception					−0.150	0.029	−0.120
					(0.122)	(0.162)	(0.154)
Implied contract exception					−0.024	−0.038	−0.035
					(0.077)	(0.081)	(0.071)
Democratic governor					−0.061	0.003	0.002
					(0.042)	(0.031)	(0.029)
Prog. inspection rate					−0.023	−0.005	−0.011
					(0.024)	(0.016)	(0.013)
Observations	911	911	911	911	911	911	911
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State trend	No	No	No	No	No	No	Yes
Division-year FE	No	No	No	No	No	Yes	Yes

The dependent variable is the natural log of the number of workplace death rate divided by the state labor force (in thousands). The primary independent variables are dummy variables for whether a state has adopted the public policy exception or whistleblower or workers' compensation statutes. Regressions are weighted by state employment in the first year of the sample, with robust standard errors clustered by state in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

and show that our estimates are robust to heterogeneous treatment effects.

C. *Interpreting and Comparing the Magnitudes of our Baseline Estimates*

The estimates in section VA consistently imply that the public policy exception to at-will employment led to an economically meaningful improvement in workplace safety. Consider that, over roughly our sample period, overall fatal workplace injury and illness rates in the United States declined from 7.5 per 100,000 workers in 1980 to 4.4 per 100,000 workers in 1994, a decline of 41% (Centers for Disease Control, 1998).[18] Over this same time period, we calculate that the share of U.S. employment in a state that had adopted the public policy exception rose 45 percentage points, from 39 percent to 84%. Using the median estimate across the specifications in table 3 (which is −0.132), our results imply that adoption of the public policy exception accounts for 13.6 percent of the overall decline in fatal injury rates over this period.[19]

In contrast, we find less convincing evidence that the two statutory protections against retaliatory firing improved safety. Our estimates tend to imply that the whistleblower statutory protection led to a reduction in injury rates, but the magnitude is smaller than the public policy exception, and the estimate is more sensitive to specification choices. We find no evidence that the workers' compensation statute had any effect on safety, on average. Given the institutional detail described in sections II and appendix A.2.2, it is not necessarily surprising that the statutory protections would have a weaker average effect on safety than the public policy exception: the penalty for employers who failed to abide by the whistleblower statute was lower on average than the pubic policy exception, average penalties under the workers' compensation statute were even lower, and penalties for both varied widely across states. Consistent with these legal protections raising employers' costs of maintaining hazardous workplaces, we would expect that the protections with the largest penalties for employers—the public policy exception—would spur the largest improvement in safety.

While this logic implies that the *average* safety effect of the statutory protections would be lower than the public-policy exception, it also implies that the statutes' effect should be *heterogeneous* depending on the damages available to one filing a claim. One of the most salient differences between the public policy exception and the two statutory protections is the availability of punitive damages which—unlike compensatory damages or reinstatement requirements—have no upper bound and can be quite substantial. Whereas the public policy exception uniformly allowed a plaintiff to sue for punitive damages, only a handful of statutory protections included them.

We test for such heterogeneity in appendix C.5. The results imply that statutes that include punitive damages do indeed reduce workplace injuries more than statutes without them.

VI. Why Does Legal Protection against Retaliation Improve Safety?

The above results reveal that legal protection against retaliatory firing for filing workers' compensation or whistleblowing substantially improved workplace safety. The discussion in section IIA outlined one mechanism that could

[18]Note that while the sample period in our regressions includes 1979–2005, the 1980–1994 span of the data from the Centers for Disease Control (1998) was the best overlap we could find for panel data on fatal workplace injuries in the United States. One reason we cannot include years beyond 1994 is that in 1995, the system for measuring fatal workplace injuries changed from the National Traumatic Occupational Fatalities surveillance system, under the National Institute for Occupational Safety and Health, to the Bureau of Labor Statistics Census of Fatal Occupational Injuries. Because these two systems used different methods, their numbers are not directly comparable.

[19]The calculation is $-0.124 * 0.45/(-0.41) = 13.6\%$, where $-0.124 = \exp(-0.132) - 1$.

drive this effect: by raising employers' expected costs of workplace injuries, such legal protections encourage employers to increase inputs into safety. The results we have shown are consistent with this mechanism. Statutory protections only improved safety when they enabled employees to sue for punitive damages (that is, when employers' penalty for violating them was sufficiently strong). The public policy exception to at-will employment, which provided a clear, uniform course of action for employees to sue for punitive damages, led to economically meaningful and persistent improvements in workplace safety.

We further examine whether this mechanism plausibly drives our results. We first consider an alternative explanation: that this relationship merely reflects changes in worker composition. We then test whether the public policy exception affected a direct measure of employers' inputs into safety. Finally, we conduct a suggestive test of whether the exception led to a larger improvement in safety when workers would be more likely to take advantage of it. We focus this analysis on the public policy exception, since the evidence in section VA revealed that the effect of the two statutory protections was low on average and highly heterogeneous.

A. *Did the Public Policy Exception Change the Workforce Composition?*

If the public policy exception is more costly for firms in hazardous sectors, then adoption of the public policy exception might lead to fewer injuries not because workplaces become safer but because workers move to safer sectors. We test for such compositional changes in table D1. We mimic table 2, except that our dependent variable is the share of state-level annual employment in manufacturing. We see no evidence that the public policy exception shifted the composition of employment in or out of manufacturing. The point estimates on the public policy exception are all tiny in magnitude, mostly insignificant, and inconsistently signed across specifications. It appears unlikely that shifts in worker composition across sectors meaningfully influence our results.[20]

B. *The Effect of the Public Policy Exception on Compliance with Safety Regulations*

By reducing workers' risks to filing for workers' compensation or blowing the whistle on their employer's safety hazards, we argue that the public policy exception would have incentivized employers to increase their inputs into safety. One salient example of employers' inputs into safety is their compliance with government safety and health regulations. Thus we examine whether noncompliance with OSHA regulations decreased following the adoption of the public policy exception. We estimate a modified version of the regression model in equation (1):

$$Noncompliance_{jsit} = \beta_1 PPE_{st} + \delta_s + \varphi_s I_s \cdot t + \tau_{dt} + \phi_{it} + X_{jsit}\gamma + \epsilon_{jsit} \quad (3)$$

Here, our dependent variable is a measure of noncompliance detected during an inspection of workplace j in state s, in industry i, conducted in year t. PPE and the sets of fixed effects are as described above, except that we also include industry-year fixed effects (ϕ_{it}) to account for time-varying shocks across industries.[21] We control for a vector of workplace-specific controls in X, including the log number of employees reported present during the inspection and if the workers were represented by a labor union, as well as time-varying state-level controls such as adoption of the statutory protections and other exceptions to at-will employment.

We quantify noncompliance as the sum of the "gravity" assigned to all violations detected in an inspection. OSHA assigns a "gravity" score to each violation based on the perceived severity of the injury that could result from the violation and the probability that such an injury would occur; the gravity of a violation ranges from 0 to 10, with 10 being most severe. We preserve the IMIS dataset at the inspection level and calculate the total gravity across all violations assessed to each inspection. Because the total gravity exhibits substantial skew but also contains a fairly large number of zeroes, we use its inverse hyperbolic sine transformation.[22]

Because violations (and associated gravity) are only observed conditional on an inspection occurring, we restrict the estimation of equation (3) to programmed inspections. As described in section IIIB, the occurrence of programmed inspections is exogenous to events at a particular workplace, conditional on industry and location; results in appendix C.2 also showed their occurrence was orthogonal to adoption of the public policy exception. In our main regressions, we additionally drop inspections in the construction sector. We do this because construction makes up the vast majority of programmed inspections (over two-thirds), and this industry disproportionately uses short-term work contracts that make wrongful discharge laws less applicable. Transactions between construction firms and contractors are typically arms-length, short-term arrangements (Cox & Thompson, 1997), and likely as a result construction workers are more likely to be contingent or alternative work arrangements[23] in which wrongful discharge laws are not applicable.

[20] A related issue is the public policy exception might lead to losses in employment for particular groups of workers with higher ex ante risk of injury, for example by raising employment costs of production workers. However, prior studies have found little to no effect of the public policy exception on employment levels or flows (Autor et al., 2007).

[21] We operationalize "industry" as 1-digit SIC industry codes. Our estimates are not materially changed by using other aggregations of industry codes.

[22] We obtain similar results using other noncompliance measures, such as the number of violations.

[23] In all years, the BLS Contingent and Alternative Employer Arrangements Survey reveals that the use of contingent work is more prevalent in construction than in other industries. See, for example, the 1999 survey: https://www.bls.gov/news.release/history/conemp_12211999.txt.

TABLE 4.—THE EFFECTS OF THE PUBLIC POLICY EXCEPTION ON COMPLIANCE WITH OSHA REGULATIONS

	(1)	(2)	(3)	(4)
Pubic policy exception	−0.042	−0.050	−0.108*	−0.145**
	(0.063)	(0.067)	(0.061)	(0.065)
Whistleblower statute		−0.074	−0.046	−0.133
		(0.079)	(0.076)	(0.084)
Workers' comp statute		0.099	−0.013	0.007
		(0.106)	(0.070)	(0.065)
Good faith exception		0.069	−0.006	0.020
		(0.118)	(0.105)	(0.052)
Implied contract exception		0.073	0.080	−0.078
		(0.049)	(0.050)	(0.059)
Log # workers present	0.040***	0.040***	0.039***	0.038***
	(0.010)	(0.010)	(0.009)	(0.009)
Union present	0.032**	0.031**	0.036***	0.037***
	(0.013)	(0.013)	(0.011)	(0.011)
Observations	317,250	317,250	317,250	317,250
Mean Dep Var (in levels)	0.808	0.808	0.808	0.808
Sector-state and sector-year FE	Yes	Yes	Yes	Yes
Industry year FE	Yes	Yes	Yes	Yes
Division-year FE	No	No	Yes	Yes
State trend	No	No	No	Yes

The dependent variable is *asinh(gravity)*, the inverse hyperbolic sine of the sum of the gravity assigned to all violations detected in an inspection, where "gravity" is a score ranging from 0–10 based on OSHA's assessment of the severity of the violation. Construction sector is omitted; see text for details. Robust standard errors clustered by state shown in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

We report our estimates without the construction sector in table 4. In column 1, we report results from a specification with state and year fixed effects only. The point estimate on the public policy exception is negative, but small in magnitude and statistically insignificant. The estimate is largely unchanged controlling for the other legal protections in column 2. When we additionally include division-year fixed effects in column 3, the point estimate increases in magnitude to −0.108 and becomes marginally significant ($p = .084$). Finally, when we additionally include state-specific trends in column 4, the estimate increases even more in magnitude and statistical significance. We estimate that adoption of the public policy exception led to a 13.5% ($\hat{\beta} = -0.145$, $p = 0.03$) reduction in the gravity assessed among nonconstruction inspections. We report results in table D2 from regressions that include the construction sector; the estimated effect of the public policy exception is qualitatively similar but, as expected, attenuates in magnitude and significance.[24]

Compliance with government safety regulations is a salient example of employers' inputs into workplace safety. The evidence in this section thus supports the hypothesis that the public policy exception increased employers' inputs into safety, which might have led to fewer workplace injuries.

C. *Heterogeneous Effects of the Public Policy Exception: The Role of Labor Unions*

We have argued that the public policy exception raised employers' expected costs of workplace injuries due to limitations in existing policies. As described in section IIA, barriers and fear of retaliation limit workers' ability to take advantage of the workers' compensation system and to blow the whistle about safety hazards. By providing protection to workers against retaliation for taking these actions, a forward-looking employer would realize that future injuries will come with higher expected costs. Such an employer has more incentive to make investments to reduce the likelihood of future injuries.

However, the extent to which legal protection against retaliation would raise employers' costs of injuries is likely to be unevenly distributed. All else equal, a change in legal protection will have a larger impact on employers' expected costs of injuries when (a) workers face lower additional barriers to filing for workers' compensation or blowing the whistle, and (b) workers are more likely to become aware of the legal change.

Both of these conditions are more likely to be true for workers represented by a labor union. Unions lower the costs of acquiring information about how to file for workers' compensation, as well as about policies that affect the benefits and costs of doing so; such information might be too costly for individual workers to acquire themselves (Weil, 1996). Unionized workers are more likely to know how to obtain workers' compensation than nonunion workers, and unions help workers through the often complex process of filing a claim (Hirsch et al., 1997; Thomason & Pozzebon, 2002). Unionized workers are more responsive to policies that affect the benefits from filing for workers' compensation such as waiting periods and benefit levels (Hirsch et al., 1997). In a similar vein, unions lower the costs for individual workers to file complaints with OSHA, both by facilitating the process and by lowering information costs (Weil & Pyles, 2005).

More generally, unions enhance the implementation of a variety of public policies centered around working

[24]The estimates that include construction in table D2 suggest that adoption of the whistleblower statute actually led to *more* gravity of violations, but the estimates are sensitive to specification and flip sign in our main results that drop the construction sector.

conditions. Mandated health and safety committees led to larger increases in OSHA enforcement for unionized workplaces than for nonunionized workplaces (Weil, 1999), and publicizing facilities' OSHA violations led to larger improvements in compliance in areas with high union density than in areas with low density (Johnson, 2020). In other contexts, unions augment collective bargaining by calling on enforcement agencies or by directly taking up legal cases on behalf of workers (Colling, 2006). Indeed, Dickens (2002) argues that "unions are an instrument in translating statute and case law into changed employment practice." In other words, even though unions might reasonably be considered a substitute for protections codified in employment law, they actually often serve as a *complement* to them.

For all these reasons, it is plausible that legal protections against dismissal for workers' compensation filing or whistleblowing would lead to a larger increase in employers' expected costs of injuries—and thus a larger reduction in injuries—among unionized workplaces relative to nonunionized workplaces.

On the other hand, other arguments imply that legal protection against dismissal would have a *smaller* safety effect in unionized workplaces. Unions often include "just cause" provisions in their collective bargaining agreements, which already put restrictions on employers' ability to discharge workers in ways that overlap with (and subsume) the public policy exception. Given this redundancy, the public policy exception might have less potential to improve safety in unionized workplaces.

However, while just cause provisions would in theory render the public policy exception less useful for unionized workers, the historical context suggests a more nuanced interpretation. During the 1980s (the period when many states adopted the public policy exception), legal scholars considered the concept of "just cause" to be ambiguous, nebulous, and not well-understood (Abrams & Nolan, 1985). By defining one explicit condition that constituted "just cause," the public policy exception could have provided a concrete guidepost for unions to exercise the rights that their contracts in theory provided. Indeed, many unions include provisions in collective bargaining agreements to enforce federal, state, and local statutes; while seemingly redundant, such provisions are useful because it is easier for workers to enforce a contract provision through a grievance procedure than to navigate statutory or common law procedures.[25]

We examine whether union presence moderates the safety effect of the public policy exception in two ways. First, we directly test whether the public policy exception affected injury rates differentially at unionized and nonunionized workplaces. We use the variable in the OSHA IMIS dataset indicating whether a union was present during the inspection to create annual state-level measures of injury counts for unionized and nonunionized workplaces. To construct these separate injury *rates*, we divide these counts by state-level annual unionized and nonunionized employment. (We describe in section IIIC how we measure unionized and nonunionized employment.)

The second way that we test the moderating role of union presence is by examining whether the safety effect of the public policy exception differs in states that had previously enacted right-to-work (RTW) laws. These laws allow workers to decline to pay union dues even if they are covered by a collective bargaining agreement, leading to free-rider problems. They have been shown to decrease union organizing (Eren & Ozbeklik, 2016), decrease union membership (Ellwood & Fine, 1987), and limit unions' bargaining strength (Ichniowski & Zax, 1991). These laws are also correlated with other "probusiness" policies that disproportionately benefit employers over workers (Holmes, 1998), potentially capturing other characteristics that limit workers' ability to leverage other favorable legal changes. We identify 16 states that had adopted RTW laws by 1979 using data from National Conference of State Legislatures (n.d.). Table D3 displays summary statistics of our primary dependent and independent variables separately for RTW and non-RTW states.

We present the results in table 5. Column 1 reports estimates from our simplest model with state and year fixed effects, similar to the estimate in column 1 of table 2, with two important changes. First, the unit of observation is now at the *state–year–union-status*, where "union-status" equals either *unionized* or *nonunionized* workplace. Second, we include an interaction of the public policy exception variable with an indicator equal to 1 for unionized workplaces. Third, because unionized and nonunionized workplaces differ in unobservable ways and may exhibit differential trends, we interact each of our set of fixed effects with a dummy for unionized workplaces.[26] The main effect of the public policy exception implies that the exception led to fewer injuries at nonunionized workplaces, but the estimate is statistically insignificant ($p = 0.13$). The interaction term (-0.101, $p = 0.047$) reveals that the public policy exception led to a substantially larger reduction in injuries among unionized workplaces; combining the two implies a safety effect for unionized workplaces of 20 percent ($-0.127 + -0.101 = -0.228$, $\exp(-0.228) - 1 = -0.203$) ($p = 0.049$).

In columns 2–4, we include region-year fixed effects (column 2), state-specific trends (column 3), and additional controls (column 4). The moderating role of unions persists, and with state-specific trends actually increases: in column 4, the main effect for nonunion workplaces is essentially zero ($\hat{\beta} = -0.011$, $p = .87$), and the interaction term increases in magnitude to -0.251, $p = .017$).

[25]For example, the faculty labor union at Rutgers University required that the university provide lactation spaces "in accordance with the law." See https://www.rutgersaaup.org/wp-content/uploads/securepdfs/2020/05/AAUP-AFT-FT-Agreement-2018-2022.pdf.

[26]Including these fixed effects without the interaction does not qualitatively change the point estimate, but—unsurprisingly—substantially increases the standard error.

TABLE 5.—HETEROGENEOUS EFFECTS OF THE PUBLIC POLICY EXCEPTION ON WORKPLACE INJURIES, BASED ON UNION PRESENCE

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Public policy exception	−0.127	−0.069*	−0.042	−0.011	0.047	0.025	0.129	0.147
	(0.078)	(0.037)	(0.064)	(0.068)	(0.094)	(0.081)	(0.111)	(0.122)
Unionized workplaces = 1 × Public policy exception	−0.101**	−0.081*	−0.251**	−0.251**				
	(0.050)	(0.047)	(0.102)	(0.102)				
No Right-to-Work law = 1 × Public policy exception					−0.245**	−0.186*	−0.323**	−0.325**
					(0.111)	(0.102)	(0.148)	(0.139)
Prog. inspection rate				−0.018				−0.013
				(0.041)				(0.041)
Whistleblower statute				−0.092				−0.108
				(0.067)				(0.069)
Workers' comp statute				0.096				0.040
				(0.122)				(0.129)
Observations	2,154	2,154	2,154	2,154	1,050	1,050	1,050	1,050
Mean Dep Var	−3.483	−3.483	−3.483	−3.483	−3.746	−3.746	−3.746	−3.746
State and year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Division-year FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
State trend	No	No	Yes	Yes	No	No	Yes	Yes
Additional Controls	No	No	No	Yes	No	No	No	Yes

In columns 1–4, the unit of observation is a state-union-status–year (where "union status" is a dummy variable that distinguishes unionized and nonunionized workplaces). We measure the injury rate at unionized workplaces as the natural log of the number of OSHA injury inspections in which a labor union was present, divided by the number of unionized workers in the state-year in thousands. We measure the injury rate at nonunionized workplaces analogously. In columns 5–8, *No RTW law* is a dummy equal to 1 if a state had not adopted right-to-work (RTW) laws as of 1979. In columns 1–4, each of the fixed effects noted in the table footer is interacted with the dummy for workplace unionization. The specifications in columns 5–8 also controls for year fixed effects interacted with the no-RTW-law dummy. Columns 4 and 8 include the same controls as in column 7 of table 2. Regressions are weighted by state employment in the first year of the sample, with robust standard errors clustered by state in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

In columns 5–8, we instead interact the public policy exception with our other measure of union presence: an indicator that a state had *not* passed RTW laws before 1979. Since states with RTW laws differ from those without RTW laws, we also include year fixed effects interacted with a no-RTW-law dummy.[27] This model implies a similar moderating effect of union presence as columns 1–4. Across all columns, the main effect of the public policy exception implies that the safety effect of the public policy exception was, if anything, *positive* in RTW states, though the estimate is always small and nowhere near statistically significant. On the other hand, in all specifications, the interaction term reveals that the public policy exception's effect on injuries is much more negative in non-RTW states. Considering the most saturated model in column 8, combining the main effect and interaction terms implies a safety effect in non-RTW states of 20 percent ($0.147 + -0.325 = -0.222$), $\exp(-0.222) - 1 = -0.20$ ($p < .01$).

While the public policy exception in principle protected all workers from retaliation, it would have raised employers' costs of injuries only if it actually made workers more inclined to file for workers' compensation or blow the whistle. Unionized workers would arguably be more responsive to a legal change that lowered the costs of taking these actions, making it plausible that the public policy exception raised employers' costs of injuries more in unionized workplaces. The results in this section support this premise. These results do not definitively establish the mechanism through which the public policy exception improved workplace safety, but they are consistent with the hypothesis that the exception improved safety by raising (some) employers' implicit costs of injuries.

[27]This inclusion has essentially no effect on the estimates.

VII. Conclusion

Laws and public policies exist—through the workers' compensation system and government safety regulations—to ensure that employers face incentives to limit workplace injuries and illnesses. However, at-will employment relationships were historically effectively treated as outside the bounds of these laws. This changed with the adoption of various legal protections that forbade employers from retaliating against workers for filing a workers' compensation claim or reporting illegal conditions. We found that states' adoption of the public policy exception to at-will employment—the most salient such legal protection—led to sustained and economically meaningful reductions in workplace injuries. Statutory protections for these actions, which tended to offer smaller damages for claimants, had a smaller effect on safety. These results imply that employers might not have been internalizing the costs of workplace injuries as existing policies had intended.

An important caveat is that we cannot definitively say that the safety improvements spurred by the public policy exception raised welfare. However, given the enormous economic burden of workplace injuries, the social benefits of the reduction in injuries were undoubtedly substantial. Additionally, no existing evidence suggests that the public policy exception meaningfully raised employers' costs: the public policy exception had no effect on employment (MacLeod & Nakavachara, 2007) or on establishment entry or productivity (Autor et al., 2007). It thus seems unlikely that safety improvements spurred by the public policy exception were welfare diminishing. Moreover, our results reveal that a sentiment embodied by prior literature, that "the public policy exception is not generally thought to impose substantial constraints on employer behavior" (Autor et al., 2007,

pg. F192), overlooked an important element of employer behavior: investments in workplace safety.

Our results inform understanding of how safety is provided in the labor market. If the level of workplace injury risk is determined in a frictionless, competitive labor market, then employers fully internalize the costs of injuries via the wage premiums they must pay workers to accept risky jobs. In this scenario, protections against dismissal for filing a workers' compensation claim or for whistleblowing have little scope to improve safety; furthermore, any improvements in safety would impose large costs on firms and/or workers. However, imperfect competition in labor markets—which much research reveals is pervasive—attenuates this market discipline. Additionally, theoretical arguments that at-will employment leads firms to underinvest in relationship-specific assets like workplace safety imply that injuries might be inefficiently high absent dismissal protections. These departures from perfect competition illustrate how the legal protections against dismissal that we study could improve safety without meaningfully affecting firms' costs.

While we find that the public policy exception adopted via the courts improved workplace safety, statutory protections with similar intents on average had either a smaller effect (whistleblower protections) or no measurable effect (workers' compensation protections). We argue that the disparity between common-law and statutory protections is not surprising, given the variance in remedies available from statutes, which were always equal to or lesser than those available under court-adopted equivalents. Furthermore, remedies available through the courts add an additional party—namely lawyers—that is motivated to encourage and inform workers regarding the process of seeking redress, increasing the likelihood of employees exercising such rights. Supporting this rationale, we find that statutes with harsher penalties did in fact deter a significant number of injuries.

At the same time, we found that the public policy exception appeared to improve safety to a much greater extent in unionized workplaces, or among workplaces located in states without policies that diminish unions' power. Given the many sources of support that unions provide workers to file for workers' compensation or complain to OSHA, nonunionized workers may have not had the support, information, or power to leverage the legal protections against unjust dismissal.

The role of union presence might explain why today—even after 43 states have adopted the public policy exception—there remain widespread reports of employers retaliating against workers for actions like filing for workers' compensation (Bernhardt et al., 2009). Given that the share of workers covered by a labor union is much lower today than it was in our sample period comprising the 1980s and 1990s, fewer workers today might be able to take advantage of legal protections offered by the public policy exception than when these laws were initially passed.

REFERENCES

Abrams, Roger I., and Dennis R. Nolan, "Toward a Theory of 'Just Cause' in Employee Discipline Cases," *Duke Law Journal* 1985 (1985), 594–623. 10.2307/1372372

Acharya, Viral V., Ramin P. Baghai, and Krishnamurthy V. Subramanian, "Wrongful Discharge Laws and Innovation," *The Review of Financial Studies* 27:1 (2013), 301–346. 10.1093/rfs/hht009

Allen, Marshall, "A Nurse Bought Protective Supplies for Her Colleagues Using GoFundMe. The Hospital Suspended Her," ProPublica (2020), https://www.propublica.org/article/a-nurse-bought-protective-supplies-for-her-colleagues-using-gofundme-the-hospital-suspended-her.

Altman, Leslie M., John M. Cerilli, Jr. Eduardo Cuaderes, Natalie C. Gros, Harry D. Jones, Bonnie L. Kristan, Patrick H. Lewis, Eugene Ryu, Michael N. Salveson, Michael T. Short, and Kimberly A. Zabroski, "Littler's Workers' Compensation Retaliation Survey," Littler's technical report (2012).

Autor, David H., John J. Donohue III, and Stewart J. Schwab, "The Costs of Wrongful-Discharge Laws," this REVIEW 88:2 (2006), 211–231.

Autor, David H., William R. Kerr, and Adriana D. Kugler, "Does Employment Protection Reduce Productivity? Evidence from US States," *The Economic Journal* 117:521 (2007), F189–F217.

Azar, José, Ioana Marinescu, and Marshall Steinbaum, "Labor Market Concentration," *Journal of Human Resources* (2020), 1218–9914R1.

Bernhardt, Annette, Ruth Milkman, Nik Theodore, Douglas D. Heckathorn, Mirabai Auer, James DeFilippis, Ana Luz González, Victor Narro, and Jason Perelshteyn, "Broken Laws, Unprotected Workers: Violations of Employment and Labor Laws in America's Cities," technical report (2009).

Biddle, Jeff, and Karen Roberts, "Claiming Behavior in Workers' Compensation," *Journal of Risk and Insurance* 70:4 (2003), 759–780. 10.1046/j.0022-4367.2003.00074.x

Bird, Robert C., and Donald J. Smythe, "The Structure of American Legal Institutions and the Diffusion of Wrongful-Discharge Laws, 1978–1999," *Law & Society Review* 42:4 (2008), 833–864.

Boone, Jan, Jan C. Van Ours, Jean-Philippe Wuellrich, and Josef Zweimüller, "Recessions Are Bad for Workplace Safety," *Journal of Health Economics* 30:4 (2011), 764–773. 10.1016/j.jhealeco.2011.05.013

Bruce, Christopher J., and Frank J. Atkins, "Efficiency Effects of Premium-Setting Regimes under Workers' Compensation: Canada and the United States," *Journal of Labor Economics* 11:1, Part 2 (1993), S38–S69. 10.1086/298328

Callahan, Elletta S., and Terry M. Dworkin, "The State of State Whistleblower Protection," *American Business Law Journal* 88 (2000), 99. 10.1111/j.1744-1714.2000.tb00286.x

Callaway, Brantly, and Pedro H. C. Sant'Anna, "Difference-in-Differences with Multiple Time Periods," *Journal of Econometrics* 225:2 (2021), 200–230. 10.1016/j.jeconom.2020.12.001

Carville, Olivia, Emma Court, and Kristen Brown, "Hospitals Tell Doctors They'll Be Fired if They Speak Up about Lack of Gear," Bloomberg (2020), https://www.bloomberg.com/news/articles/2020-03-31/hospitals-tell-doctors-they-ll-be-fired-if-they-talk-to-press.

Centers for Disease Control, "Morbidity and Mortality Weekly Report," 47(15) (April 24, 1998), 297–302, https://www.cdc.gov/mmwr/preview/mmwrhtml/00052214.htm.

Colling, Trevor, "What Space for Unions on the Floor of Rights? Trade Unions and the Enforcement of Statutory Individual Employment Rights," *Industrial Law Journal* 35:2 (2006), 140–160. 10.1093/indlaw/dwl011

Conlon, John J., Laura Pilossoph, Matthew Wiswall, and Basit Zafar, "Labor Market Search with Imperfect Information and Learning," NBER technical report 24988 (2018).

Cox, Andrew, and Ian Thompson, "'Fit for Purpose' Contractual Relations: Determining a Theoretical Framework for Construction Projects," *European Journal of Purchasing & Supply Management* 3:3 (1997), 127–135.

Day, Matt, "Amazon Vice President Quits in Protest over Company's Alleged Firings of Coronavirus Whistleblowers," Time (May 2020). https://time.com/5831674/amazon-engineer-quits-coronavirus-whistleblowers/.

de Chaisemartin, Clément, and Xavier d'Haultfoeuille, "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects," *American Economic Review* 110:5 (2020), 2964–2996. 10.1257/aer.20181169

DeNicco, James P., "Employment-at-Will Exceptions and Jobless Recovery," *Journal of Macroeconomics* 45 (2015), 245–257. 10.1016/j.jmacro.2015.05.003

Dickens, Linda, "Individual Statutory Employment Rights since 1997: Constrained Expansion," *Employee Relations* 24:6 (2002), 619–637. 10.1108/01425450210453031

Dube, Arindrajit, Alan Manning, and Suresh Naidu, "Monopsony and Employer Mis-Optimization Explain Why Wages Bunch at Round Numbers," NBER technical report 24991 (2018).

Dube, Arindrajit, Jeff Jacobs, Suresh Naidu, and Siddharth Suri, "Monopsony in Online Labor Markets," *American Economic Review: Insights* 2:1 (2020), 33–46. 10.1257/aeri.20180150

Edelman, Lauren B., Steven E. Abraham, and Howard S. Erlanger, "Professional Construction of Law: The Inflated Threat of Wrongful Discharge," *Law and Society Review* 26 (1992), 47–83. 10.2307/3053836

Edisis, Adrienne T., "Workers Compensation Claims by Temporary Help Services Workers and Traditional Employees," unpublished manuscript (2017).

Ellwood, David T., and Glenn Fine, "The Impact of Right-to-Work Laws on Union Organizing," *Journal of Political Economy* 95:2 (1987), 250–273. 10.1086/261454

Eren, Ozkan, and Serkan Ozbeklik, "What do Right-to-Work Laws Do? Evidence from a Synthetic Control Method Analysis," *Journal of Policy Analysis and Management* 35:1 (2016), 173–194. 10.1002/pam.21861

Fan, Z. J., David K. Bonauto, Michael P. Foley, and Barbara A. Silverstein, "Underreporting of Work-Related Injury or Illness to Workers' Compensation: Individual and Industry Factors," *Journal of Occupational and Environmental Medicine* 48:9 (2006), 914–922. 10.1097/01.jom.0000226253.54138.1e

Fishback, Price V., "Liability Rules and Accident Prevention in the Workplace: Empirical Evidence from the Early Twentieth Century," *The Journal of Legal Studies* 16:2 (1987), 305–328. 10.1086/467832

Goodman-Bacon, Andrew, "Difference-in-Differences with Variation in Treatment Timing," *Journal of Econometrics* 225:2 (2021), 254–277. 10.1016/j.jeconom.2021.03.014

Government Accountability Office, "Inspectors' Opinions on Improving OSHA's Effectiveness," technical report (1990).

Groenewold, Matthew R., and Sherry L. Baron, "The Proportion of Work-Related Emergency Department Visits Not Expected to Be Paid by Workers' Compensation: Implications for Occupational Health Surveillance, Research, Policy, and Health Equity," *Health Services Research* 48:6pt1 (2013), 1939–1959. 10.1111/1475-6773.12066

Hirsch, Barry T., and David A. Macpherson, "Union Membership and Coverage Database from the Current Population Survey: Note," *ILR Review* 56:2 (2003), 349–354. 10.1177/001979390305600208

Hirsch, Barry T., and David A. Macpherson, and J. M. DuMond, "Workers' Compensation Recipiency in Union and Nonunion Workplaces," *ILR Review* 50:2 (1997), 213–236.

Holmes, Thomas J., "The Effect of State Policies on the Location of Manufacturing: Evidence from State Borders," *Journal of Political Economy* 106:4 (1998), 667–705. 10.1086/250026

Ichniowski, Casey, and Jeffrey S. Zax, "Right-to-Work Laws, Free Riders, and Unionization in the Local Public Sector," *Journal of Labor Economics* 9:3 (1991), 255–275. 10.1086/298268

Johnson, Matthew S., "Regulation by Shaming: Deterrence Effects of Publicizing Violations of Workplace Safety and Health Laws," *American Economic Review* 110:6 (2020), 1866–1904. 10.1257/aer.20180501

Kniesner, Thomas J., and John D. Leeth, "Regulating Occupational and Product Risks" (pp. 493–600), in *Handbook of the Economics of Risk and Uncertainty*, Vol. 1 (Elsevier, 2014). 10.1016/B978-0-444-53685-3.00009-X

Kniesner, Thomas J., W. K. Viscusi, Christopher Woock, and James P. Ziliak, "The Value of a Statistical Life: Evidence from Panel Data," this REVIEW 94:1 (2012), 74–87.

Krueger, Alan B., "Incentive Effects of Workers' Compensation Insurance," *Journal of Public Economics* 41:1 (1990), 73–99. 10.1016/0047-2727(92)90057-M

Lamadon, Thibaut, Magne Mogstad, and Bradley Setzler, "Imperfect Competition, Compensating Differentials, and Rent Sharing in the US Labor Market," *American Economic Review* 112:1 (2022), 169–212. 10.1257/aer.20190790

Lavetti, Kurt, "The Estimation of Compensating Wage Differentials: Lessons from the Deadliest Catch," *Journal of Business & Economic Statistics* 38:1 (2020), 165–182.

Lavetti, Kurt, and Ian M. Schmutte, "Estimating Compensating Wage Differentials with Endogenous Job Mobility," working paper (2018).

Lee, Jonathan M., and Laura O. Taylor, "Randomized Safety Inspections and Risk Exposure on the Job: Quasi-Experimental Estimates of the Value of a Statistical Life," *American Economic Journal: Economic Policy* 11:4 (2019), 350–374. 10.1257/pol.20150024

Leigh, J. P., "Economic Burden of Occupational Injury and Illness in the United States," *The Milbank Quarterly* 89:4 (2011), 728–772. 10.1111/j.1468-0009.2011.00648.x

MacLeod, W. B., and Voraprapa Nakavachara, "Can Wrongful Discharge Law Enhance Employment?" *The Economic Journal* 117:521 (2007), F218–F278.

Manning, Alan, "Imperfect Competition in the Labor Market" (pp. 973–1041), in *Handbook of Labor Economics*, Vol. 4 (Amsterdam: Elsevier, 2011). 10.1016/S0169-7218(11)02409-9

Moore, Michael J., and W. K. Viscusi, "Promoting Safety through Workers' Compensation: The Efficacy and Net Wage Costs of Injury Insurance," *The RAND Journal of Economics* (1989), 499–515.

Morantz, Alison D., "Coal Mine Safety: Do Unions Make a Difference?" *ILR Review* 66:1 (2013), 88–116. 10.1177/001979391306600104

National Conference of State Legislatures, "Right-to-Work Resources," (n.d.), http://www.ncsl.org/research/labor-and-employment/right-to-work-laws-and-bills.aspx, accessed March 2018.

OSHA, "About OSHA" (2020), https://www.osha.gov/aboutosha, accessed July 28, 2020.

Prager, Elena, and Matt Schmitt, "Employer Consolidation and Wages: Evidence from Hospitals," *American Economic Review* 111:2 (2021), 397–427. 10.1257/aer.20190690

Rosen, Sherwin, "The Theory of Equalizing Differences" (pp. 641–692), in *Handbook of Labor Economics*, Vol. 1 (Amsterdam: Elsevier, 1986). 10.1016/S1573-4463(86)01015-5

Ruser, John, and Richard Butler, "The Economics of Occupational Safety and Health," *Foundations and Trends® in Microeconomics* 5:5 (2010), 301–354.

Ruser, John W., "Workers' Compensation Insurance, Experience-Rating, and Occupational Injuries," *The RAND Journal of Economics* 16:4 (1985), 487–503. 10.2307/2555508

——— "Workers' Compensation and Occupational Injuries and Illnesses," *Journal of Labor Economics* 9:4 (1991), 325–350. 10.1086/298272

Shannon, Harry S., and Graham S. Lowe, "How Many Injured Workers Do Not File Claims for Workers Compensation Benefits?" *American Journal of Industrial Medicine* 42:6 (2002), 467–473. 10.1002/ajim.10142

Shields, Joseph, Xiaohua Lu, and Gaylon Oswalt, "Workers' Compensation Deductibles and Employers' Costs," *Journal of Risk and Insurance* 66 (1999), 207–218. 10.2307/253609

Sinzdak, Gerard, "An Analysis of Current Whistleblower Laws: Defending a More Flexible Approach to Reporting Requirements," *California Law Review* 96 (2008), 1633.

Spieler, Emily A., and John F. Burton Jr., "The Lack of Correspondence Between Work-Related Disability and Receipt of Workers' Compensation Benefits," *American Journal of Industrial Medicine* 55:6 (2012), 487–505. 10.1002/ajim.21034

Thomason, Terry, and Silvana Pozzebon, "Determinants of Firm Workplace Health and Safety Claims Management Practices," *Industrial and Labor Relations Review* 55:2 (2002), 286–307. 10.1177/001979390205500205

Viscusi, W. K., and Joseph E. Aldy, "The Value of a Statistical Life: A Critical Review of Market Estimates throughout the World," *Journal of Risk and Uncertainty* 27:1 (2003), 5–76. 10.1023/A:1025598106257

Viscusi, W. K., and Michael J. Moore, "Worker Learning and Compensating Differentials," *ILR Review* 45:1 (1991), 80–96. 10.1177/001979399104500106

Weatherford, Katie, "Securing the Right to a Safe and Healthy Workplace: Improve State Laws to Protect Workers," *Center for Effective Government* (2013).

Weil, David, "Regulating the Workplace: The Vexing Problem of Implementation," *Advances in Industrial and Labor Relations* 7 (1996), 247–286.

——— "Are Mandated Health and Safety Committees Substitutes for or Supplements to Labor Unions?" *ILR Review* 52:3 (1999), 339–360. 10.1177/001979399905200301

Weil, David, and Amanda Pyles, "Why Complain—Complaints, Compliance, and the Problem of Enforcement in the US Workplace," *Comparative Labor Law & Policy Journal* 27 (2005), 59.

Williamson, Oliver E., Michael L. Wachter, and Jeffrey E. Harris, "Understanding the Employment Relation: The Analysis of Idiosyncratic Exchange," *The Bell Journal of Economics* 6:1 (1975), 250–278. 10.2307/3003224

Young, Alwyn, "Channeling Fisher: Randomization Tests and the Statistical Insignificance of Seemingly Significant Experimental Results," *The Quarterly Journal of Economics* 134:2 (2019), 557–598. 10.1093/qje/qjy029

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